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1. Abstract:

Air Pollution is not just a simple issue; it is a multidimensional environmental and global crisis. In rapidly evolving economies like India, the intersection of urbanization and industrialization creates a stochastic environment in which air quality degradation is not merely a steady decline but a series of complex fluctuations. It has become crucial not only to monitor air quality but also to understand and predict its future behavior.

One of the critical challenges faced are the inconsistency and complexity of the pollution sources. Pollution is heavily influenced by a combination of factors that may vary depending upon the location and conditions. Meteorological conditions during winter and post-monsoon periods create unique "pollution signatures" that are difficult to categorize using traditional statistical methods.

All these challenges motivate the need for a data driven approach focusing on both understanding need for a robust, data-driven approach that transitions from mere monitoring to predictive intelligence. By implementing classical supervised machine learning models; specifically Multinomial Logistic Regression, Support Vector Machines, and Random Forest we seek to effectively classify seasonal air quality levels. The integration of these techniques ensures high predictive accuracy while maintaining the interpretability and relevance necessary for understanding complex environmental data.

2. Introduction:

The foundational step for the project was to secure a robust dataset. This turned out to surprisingly difficult, as most of the datasets were either feature-poor or contained several missing values. After carefully assessing the datasets, a dataset with 8 relevant features was selected. This dataset did not have any missing data. However, it consisted of some inconsistent values that were removed during the preprocessing, to ensure the quality of the data.

The core objective of this project was to perform a comparative analysis amongst classical supervised machine learning models for predicting AQI categories based on the relevant parameters. Through this project we also try to lay emphasis on the importance of traditional ML models, which have been overshadowed by the increasing popularity of deep learning models.

Additionally, we try to demonstrate the importance of model selection and training in order to obtain accurate results. By training and comparing the three model, we aim towards identifying the most suitable model for AQI classification.

The methodology and findings of this study will also be connected to existing literature review, to provide a broader context of the obtained results and supporting the relevance of the chosen approach.

3. Literature Survey:

One of the critical environmental issues caused by industrial emissions, burning of fossil fuel, vehicle exhaust and other urban activities is Air pollution. Pollution levels are measured by AQI (Air Quality Index) which helps in accurately predicting and identifying between pollutants and air quality levels.

The research paper "Optimized Machine Learning Model for Air Quality Index Prediction in Major Cities in India" [7] proposed a hybrid machine learning model combining Grey Wolf Optimization on (GWO) and Decision Tree regression for AQI prediction on across the major Indian ci es. GWO selects features of most important pollutants, whereas Decision Tree predicts AQI categories. The issue of missing values is handled by data processing. SMOTE helps in balancing data. The study reveals high prediction on accuracy (around 88-97%) indica ng that optimized machine learning models improve prediction on performance. PM2.5 and PM10 are the most important pollutants affecting the AQI levels. The strength of the study lies in accuracy improvement using optimization techniques, while the limitation on limited seasonal analysis and restricted dataset diversity.

The research paper "Seasonal Analysis of Air Pollution on in India (Coastal and Non-Coastal Ci es)" [8] focuses on how seasons change affect the pollution on levels using machine learning techniques such as LSTM (Long Short-Term Memory) are used predict AQI values and analyze pollutant trends. PM2.5 and PM10 strongly affect AQI levels across seasons. During the rainfall season, AQI levels are generally lower due to rainfall, which helps remove pollutants. Variation in the pollution is observed post-monsoon season due to crop burning and industrial emissions. Coastal areas show better air quality as compared to non-coastal areas due to sea breeze that helps in dispersing pollutants. The strength of the study is seasonal comparison of pollution patterns, while the limitation is limited feature diversity and limited model comparison.

Both the studies highlight that the pollutant concentration on pa erns, especially PM2.5 and PM10 play a major role in determining AQI categories. Seasonal variation affects the pollution levels, with post-monsoon and winter resulting in high AQI variability compared to monsoon season. These findings support the use of machine learning algorithms such as Support Vector Machine, Random Forest and regression models that help to predict AQI categories using seasonal features and pollutant concentration. This analysis provides a strong base for building accurate prediction on models and analyzing seasonal air pollution.

4. Dataset Description:

The dataset used in this study is sourced from a publicly available dataset on Kaggle titled "India City Air Quality Index Dataset (2015-2023)" [6]. It consists of AQI measurements collected from multiple cities over several years. It includes key pollutants such as PM2.5, PM10, NO2, SO2, CO and O3, in addition to AQI values and their categorical classifications. These Features represent both particulate as well as gaseous pollutants which determine the overall air quality.

Exploratory Data Analysis (EDA) was performed to understand the dataset's structure, distributions, and relationships. Initially, it included handling missing values and ensuring data consistency by validating the relationship between PM2.5 and PM10.

A Strong positive correlation between PM2.5 and PM10 was observed. This confirmed that they are the main drivers of AQI. In contrast, gaseous pollutants showed negligible correlation, which can be attributed to varying pollution sources across different regions. When combined, this leads to the Geographical Dilution Effect.

The statistical analysis indicates low skewness (approx. 0.01-0.15) and negative kurtosis. This suggests relatively symmetric distributions with occasional extreme pollution spikes.

To incorporate temporal patterns, a new feature "season" was derived from the date column (Winter, Summer, Monsoon, Post-Monsoon). Seasonal analysis shows that while average AQI remains stable, higher variability and extreme pollution events occur during winter and post-monsoon time period.

The dataset also shows class imbalance along with higher pollution categories being underrepresented. This may affect model performance. VIF analysis indicates values within the acceptable range (1-5). This suggests that there is no severe multicollinearity.

Overall, the analysis indicates that the particulate matter concentration is a stronger predicator of AQI than geographical or seasonal factors.

5. Methodology and Model Detail:

5.1. Data Acquisition

As mentioned in the introduction, we wanted to obtain a robust dataset in-order to ensure efficient model training. Most of the government institutions have several inconsistencies such as incomplete datasets, sparse features. After exploring multiple datasets, we reduced our feature set to 8 relevant features. After analyzing multiple datasets, we found a dataset that had all the relevant features without any missing values. However, there were several entries with inconsistencies, details of which were covered in the section Dataset Description.

5.2. Pre-Processing

The dataset underwent several preprocessing steps to ensure data quality and consistency. The missing values were handled using imputation techniques to maintain continuity. The date column was converted into a datetime format. This enabled extraction of temporal features such as month and season. A new feature "season" was created to capture time-based variation in air quality. Categorical variables such as AQI categories were properly encoded for modeling. Data consistency checks were performed, particularly for the relationship between PM2.5 and PM10 and extreme values were retained as they represent real world pollution events.

5.3. Model Details

We will now examine our trained models, highlighting their design and implementation.

5.3.1 Multiple Linear Regression

- **Definition:** Multiple Linear Regression (MLR) is a supervised learning algorithm that models the relationship between a single continuous target and multiple independent features. It is a parametric model, that assumes the data follows a specific linear functional form.

The methodological approach was densely mathematical. To estimate the raw numerical AQI value (y). The model followed the fundamental equation:

$$AQI = \beta_0 + \beta_1(PM_{2.5}) + \beta_2(PM_{10}) + \text{`s} + \beta_n(\text{Season_Dummy}) + \varepsilon$$

This ensured that the model captured the continuous AQI while prioritizing high-pollution "Poor" events that were under-represented in the dataset. The Weighted Regression-to-Classification Pipeline allowed the model to see the subtle gradients in pollution before forcing them into buckets.

- **Training, Feature Engineering and Scaling:**
  Stratified Splitting on "aqi_category" was performed to ensure representative class distribution in the training and test split of 80:20.
  Features were standardized to have a mean of 0 and a variance of 1.

  The Weighted Least Squares (Sample Weighting) approach was employed for gradient update.
  Higher weights (w_i) are assigned to "Poor" and "Moderate" categories to penalize errors on high-pollution days more severely.
  - Poor Category: Weights range from 500 (Monsoon) to 3000 (Post-Monsoon).
  - Moderate Category: Assigned a slight boost weight of 5.
  - Satisfactory/Good: Assigned a normal weight of 1.

- **Post-Prediction Classification and Evaluation:**
  To bridge the gap between continuous estimation and official regulatory reporting, we map as Continuous Prediction to CPCB Mapping.
  Thresholding Logic:
  - AQI ≤ 50: Good
  - 51 ≤ AQI ≤ 100: Satisfactory
  - 101 ≤ AQI ≤ 200: Moderate
  - 200 ≤ AQI: Poor

- **Visualization:**
  Comprehensive visualizations interpret model performance and feature importance.
  - **Regression Accuracy:** Scatter plots compare "Actual vs Predicted AQI" across different seasons to visualize the "Ideal Fit" line.
  - **Feature Impact:** A bar chart visualizes the Regression Weights (Coefficients), showing the relative impact of features on th, final AQI prediction.
  - **Confusion Matrix:** A heatmap illustrates the classification accuracy.

5.3.2 Support Vector Machine

- **Definition: "**A Support Vector Machine (SVM) is a supervised machine learning algorithm primarily used for classification and regression tasks that constructs a hyperplane to separate data points into different classes" [1].
- **Training:** While training the SVM model, it was important to include only the relevant features and remove those that did not have a major significance. For example, feature like date was removed as it did not have a direct impact on the classification process. Since the "Poor" AQI category only had 10 instances, it was decided to include it along with the "Moderate" category. Unlike the MLR model, applying weights to the categories did not have a significant effect on the accuracy.
  Categorical features such as city and season were one hot encoded to a numerical format suitable for the model.
- **Scaling:** Since SVM models are sensitive to feature scales, standardization was applied to all the features. The model was implemented using the RBF kernel, which effectively captures non-linear relations.
- **Evaluation:** Model evaluation was performed using the following metrics: accuracy, precision, recall, f1-score, with an emphasis on the macro f1-score to ensure a balanced contribution of all the classes.
- **Visualization:** To enhance the interpretability, we used PCA to visualize the decision boundaries and plot the features in two-dimensions.

5.3.3 Random Forest

- **Definition:** Random Forest is a supervised, ensemble machine learning algorithm used for classification and regression. It works by building up multiple decision trees and combines the final prediction based on majority vote to reduce overfitting and improve accuracy.

- **Training:** While training the random forest model, it became crucial to select the input features especially pm2.5 and pm10 as they were considered the most important feature as compared to others. Also, the "Poor" AQI category only had 10 instances, so it was decided to include it along with the "Moderate" category. Each tree learned different patterns efficiently, improving prediction accuracy.

- **Data Preprocessing:** Handling the categorical variables using encoding and ensuring numerical consistency in features are the important step under data pre-processing.

- **Evaluation:** Model evaluation was performed using the following metrics: accuracy, precision, recall, cross validation, f1 score with an emphasis on the macro-f1 to ensure a balanced contribution of all the classes.

- **Visualization:** To enhance the interpretability, we plot a feature importance graph which provides an important score for each feature. Accuracy vs Number of Trees Plot, helped to determine the optimal number of trees. Confusion Matrix Heatmap visualization helped classify performance for each AQI category.

5.3.4 Evaluation Metrics

It is seen that all models achieved very high levels of accuracy (greater than 98%), and this happened because of good feature selection and absence of missing values in the dataset. But assessing models based only on accuracy was not sufficient as all models showed high accuracy.
Thus, it became necessary to use other criteria of assessment for detailed analysis. These metrics include:

- Accuracy
- Macro F1
- Train-Test Gap
- Cross-validation
- Feature Ablation
- Cohen's kappa

The procedure of feature ablation involved eliminating the most important features, which were PM2.5 and PM10, and checking the performance of the model. A smaller drop in accuracy indicates a more robust model.

6. Results and analysis

6.1. General Observation

All the three models consistently delivered high performance (above 98%), along with strong Macro f1 score and Kappa values. This shows that the dataset is well-structured, and contains the features that are highly predictive.

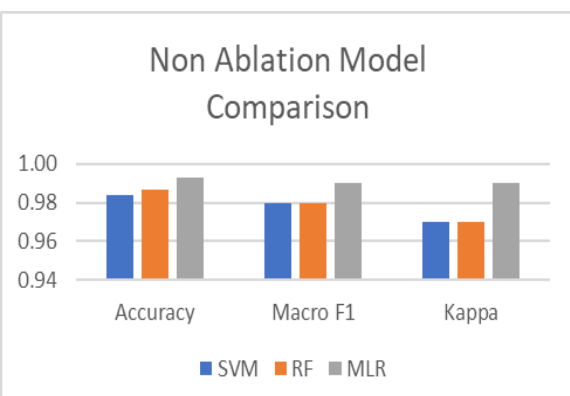
6.2. Model Comparison

Classification models like SVM and Random Forest showed stable performance with less than 1% of train and test difference and consistent cross-validation score. On the other side, the Multiple Linear Regression model illustrated a slightly higher numerical performance. This was due to its indirect approach of predicating AQI and mapping it to their respective categories.

Model	Accuracy	Macro F1	Kappa	CV Score	Train-Test Gap
SVM	98.4%	0.98	0.97	High	Low
RF	~98%	0.98	~0.97	Stable	Slight
MLR	99.3%	0.99	~0.99	Very High	Very Low

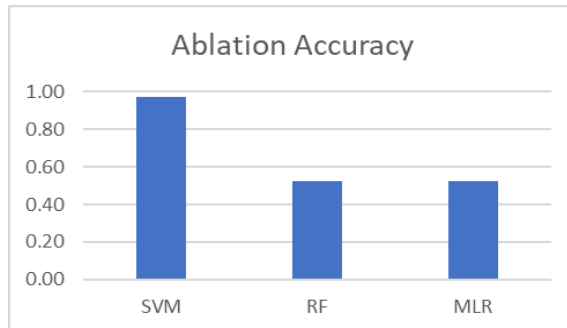
6.3. Performance Evaluation Metrics

To ensure a comprehensive and fair evaluation amongst all the models, it was decided to consider multiple performance metrics. Measures like accuracy and f1-score measured performance. The train-test gap assessed overfitting, cross-validation ensured stability and Cohen's Kappa confirmed that the performance is not by chance.



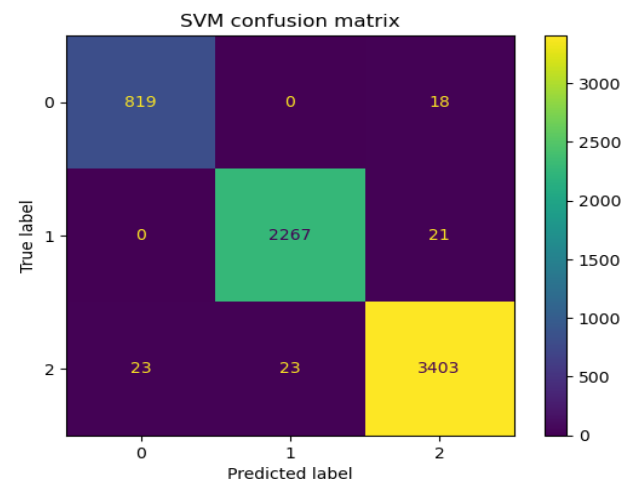
6.4. Feature Ablation

Feature Ablation was performed by removing the important features, namely PM2.5 and PM10. The SVM models showed only slight drop in performance, indicating robustness and correlated features contributing to prediction.



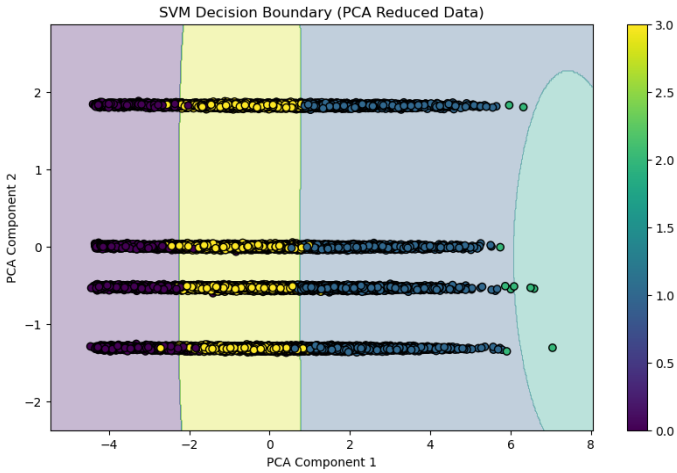
6.5. Critical Analysis

Although Multiple Linear Regression achieved the highest numerical values, it follows an indirect approach and is not suitable for classification purpose. The MLR model only fits in the scenarios where the model is required to be simple and light weight. In contrast, SVM and Random Forest directly model class boundaries and provide more meaningful results



6.6. Model Selection

Taking into consideration that MLR is working on an indirect approach of classification, SVM was selected as the best model for the AQI classification, due to its consistent performance through all the metrics. This shows its robust nature and strong generalizing ability.



7. Conclusion and Future work

7.1. Conclusion

In this project we have successfully trained and evaluated multiple Machine Learning models for AQI classification using both pollution and contextual features. All the models had shown a high performance, indicating that the dataset was well structured and predictive.

Among all the models, the Support Vector Machine (SVM) had demonstrated the best overall performance, with strong accuracy, balanced macro F1-score and Cohen's Kappa, along with minimal overfitting and robust under feature ablation.

The study also highlighted that while regression based approached can achieve high performance, they are only suitable for light weight tasks. For extreme and critical classification, direct classification models should be used. Moreover, feature ablation depicted that the SVM model did mot only depend on the individual features, indicating strong robustness and presence of correlated variables.

Overall, the results show effectiveness of machine learning models in predicting AQI categories and provide a stable framework AQI analysis.

7.2. Future Work

The current study has achieved strong results; several improvements can be incorporated in the future work. For instance, inclusion of meteorological parameters such as temperature, humidity, wind speed etc. This could further enhance the model's performance.

More advanced methods like hyperparameter optimization, ensemble learning, and deep learning may also be considered to enhance the predictive ability. Additionally, incorporating real-time data through API and other sources and implementing the model as a web/mobile application would render it more practical and convenient.

A deeper analysis with bigger and more varied datasets would allow the model to generalize better and be more applicable in real-world situations.

8. References

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